
Compiler Design

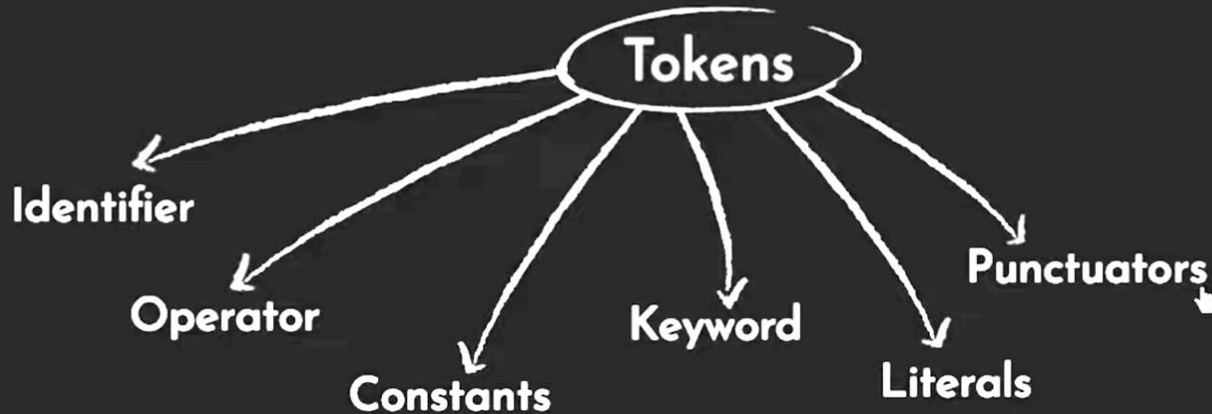
Lecture-02

Lexical Analyzer

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Course: CSE 4111

Lexical Analyzer:

- Scans the Pure HLL code line by line.
- Takes Lexemes as i/p and produces Tokens.



Tokenization

```
int main()
{
    int x,a=2,b=3,c=5;
    x = a+b*c;
    printf("The value of x is %d",x);

    return 0;
}
```

Lexical Analyzer–Tokenization:



Tokens:

1. Keyword: int , return
2. Identifier: main , x , a , b , c , printf
3. Punctuator: (,) , { , , , ; , }
4. Operator: = , + , *
5. Constant: 2 , 3 , 5 , 0
6. Literal: "The value of x is %d"

```
int main()
{
    int x,a=2,b=3,c=5;
    x = a+b*c;
    printf("The value of x is %d",x);

    return 0;
}
```

Count: 39



Lexical Error

- lexical error occurs when a sequence of characters does not match the pattern of any token.
- It typically happens during the execution of a program.
- Types of errors:
 - Exceeding length of identifier or numeric constants.
 - Appearance of illegal characters
 - Unmatched string
 - Spelling Error



Error Recovery Techniques

1. Panic Mode Recovery
2. Transpose of two adjacent characters.
3. Insert a missing character
4. Delete an unknown or extra character
5. Replace a character with another.



Thank You